Does Cell-Weighted Sensitivity Hide Hard Patient-IDs? A Pre-Registered Patient-ID–Stratified Re-Evaluation of a ResNet-50 Malaria Cell Classifier

[Team member names]
CSA431 Computer Vision — research-reproduction project
Public NIH/NLM segmented-cell dataset; no clinical claims

***Abstract*—Malaria cell classifiers on the public NIH/NLM thin-smear release are usually summarised by pooled, cell-weighted metrics. We reproduced Rajaraman et al.'s frozen ResNet-50 feature-extractor baseline under five folds grouped by the official NLM Patient-ID mapping key. The reproduction reached accuracy 93.90%, sensitivity 94.24% and AUC 0.9832, against the published 95.9%, 94.7% and 0.991. Before any analysis, we pre-registered one change: re-aggregate the frozen out-of-fold predictions per Patient-ID. We predicted that Patient-ID-macro sensitivity would fall below pooled sensitivity by about 1–3 percentage points (pp). The rule was: supported if at least 4 of 5 folds are negative and the mean is at most -1.68 pp. Observed: 4 of 5 folds were negative, with a mean difference of -1.56 pp, inside the predicted range but short of the threshold. The pre-registered verdict is therefore INCONCLUSIVE. Exploratory ablations show the result is fragile: 2 of 151 single-Patient-ID deletions would flip it. The one positive fold is explained by large Patient-IDs that are hard rather than easy. Failure analysis finds false negatives concentrated in few Patient-IDs: the top ten hold 51.4% of them. A blind, non-expert image review found that most of the model's most confident errors are cells whose visible content disagrees with their label. This is an evaluation study on a public dataset; it makes no clinical claim.**

***Index Terms*—malaria, cell classification, evaluation protocol, grouped cross-validation, sensitivity, pre-registration, failure analysis**

I. Introduction

Thin-blood-smear microscopy can be posed as a binary computer-vision task on segmented red-blood-cell crops: Parasitized vs. Uninfected. Many methods on the NIH/NLM release report 96–99% accuracy [1]–[5]. Cells are not independent, however: cells from the same source share stain, illumination and focus. Rajaraman et al. [6] therefore evaluated their pretrained-CNN feature extractors with patient-level five-fold cross-validation. Their patient-level accuracy (95.9%) is lower than their cell-level accuracy (98.6%).

Our project does not claim to discover this leakage, nor that leakage makes accuracy collapse. MalariaNet [7] reports a small slide-disjoint drop, but a clear sensitivity/specificity imbalance. The narrower question left open in our survey is how errors are distributed across source groups. Under a leakage-free protocol, does a pooled, cell-weighted mean hide groups with concentrated false negatives? We (i) reproduce the Rajaraman ResNet-50 baseline with folds grouped by the official NLM Patient-ID key; (ii) pre-register one evaluation change and its prediction; (iii) apply it to the frozen predictions with exploratory ablations; and (iv) analyse the failures. The change is a reporting change, not a model improvement.

II. Related Work

Rajaraman et al. [6] released the cell crops and compared a custom CNN with ImageNet-pretrained AlexNet, VGG-16, Xception, ResNet-50 and DenseNet-121 feature extractors. Their main protocol is patient-level five-fold cross-validation, and they also report a cell-level comparison. MalariaNet [7] repeated the comparison in 2026 across eight architectures. The slide-disjoint score was lower in every model; for MalariaNet itself, 97.06% fell to 95.61%, a gap inside its seed spread. Its leakage-free sensitivity and specificity differed (93.94% vs. 97.41%).

Several high-accuracy methods use cell-level or unstated splits. Narayanan et al. [1] use an 80/20 hold-out, and Alharbi et al. [2] a random cell-level split. EDRI [3] uses a stratified 80/10/10 cell split without k-fold. A CNN–ViT ensemble [4] reports an 80/20 split plus k-fold, but its confusion-matrix counts sum to the full 27,558 cells. ConvNeXt [5] augments 27,558 crops to 606,276 without a stated patient split. IST-Net [8] reports a second dataset; we read only its abstract and make no claim about its split. A 2024 systematic review [9] summarises the field; we read only its abstract and cite no counts from it. Wilm et al. [10] re-annotated the full smears and found many border cells unlabelled, a warning that segmented-cell labels can be noisy. None of these works reports the distribution of errors over source groups, which is the gap we test.

III. Dataset and Protocol

We use only the official NLM archive and its two Patient-ID-to-cell mapping files [11]: 27,558 PNG crops, balanced between classes. The archive and mappings match their recorded SHA-256 hashes. All 27,558 images match the per-file hashes of the audited M2 manifest. The grouping key is the exact official mapping key, called *Patient-ID* here; it is not a slide

TABLE I
PUBLISHED PATIENT-LEVEL VALUES (RAJARAMAN ET AL. TABLE 6, FROM THE PROJECT SUMMARY) VS. MEASURED M2 BASELINE (MEAN OVER FIVE PATIENT-ID-GROUPED FOLDS). PUBLISHED AND MEASURED VALUES ARE IN SEPARATE COLUMNS.

Metric	Published	Measured (M2)	Fold SD	Difference
Accuracy	95.9%	93.90%	1.95 pp	−2.00 pp
Sensitivity (recall)	94.7%	94.24%	3.36 pp	−0.46 pp
Specificity	97.2%	93.56%	1.30 pp	−3.64 pp
AUC	0.991	0.9832	0.0086	−0.0078

ID or a patient diagnosis. There are 201 Patient-IDs. 151 have Parasitized cells and 50 have only Uninfected cells, because Uninfected crops also come from infected slides. Five folds are built with `StratifiedGroupKFold` (seed 42) on these keys. The test folds hold 5,497 / 5,520 / 5,544 / 5,483 / 5,514 cells and 40 / 40 / 41 / 40 / 40 Patient-IDs, and no Patient-ID appears in both a fold's training and test sets. Parasitized is the positive class. We report accuracy, sensitivity (recall), specificity, AUC and F1 at the fixed 0.5 threshold.

IV. REPRODUCTION

Following [6], a frozen ImageNet ResNet-50 (torchvision `IMAGENET1K_V2`) extracts 2,048-d global-average-pooled features from 224×224 crops with ImageNet normalisation. A fresh 2048→512→1 classifier with ReLU and dropout 0.5 is trained per fold, using SGD (learning rate 0.01, momentum 0.9, L2 10^{-4}) for 20 epochs. These are logged approximations where the paper's details are not available to us: the selected layer, hidden width, search ranges and original fold membership. Table I compares the result with the published patient-level values. Mean sensitivity is close to the published value; accuracy and specificity are lower. This is an approximate reproduction, not a paper-exact rerun.

V. HYPOTHESIS AND PREDICTION

The hypothesis and decision rule were recorded in `milestones/M3/hypothesis.md`, dated *"3 October 2026 (Asia/Kolkata), after the saved M2 baseline and before any M4 code."* The repository history agrees: the file was first committed in `f3152be` on 03 Oct 2026 23:48 +0530, after the M2 metrics were written (2026-10-03 17:22 UTC). The record reads:

> **IF** the fixed M2 out-of-fold predictions are evaluated separately for every official Patient-ID within each unchanged patient-grouped fold, **THEN** Patient-ID-macro sensitivity for Parasitized will be lower than the existing pooled cell-weighted sensitivity […] **BECAUSE** cells from one source group share acquisition/stain characteristics and groups contribute unequal numbers of cells.

The predicted mean paired difference was "approximately −1 to −3 percentage points." The M3 decision rule had three outcomes. *Supported:* at least 4 of 5 differences negative and mean ≤ −1.68 pp, half of M2's 3.36 pp sensitivity fold SD. *Unsupported:* the mirror rule. *Inconclusive:* anything else.

VI. METHOD: PATIENT-ID–STRATIFIED RE-AGGREGATION

The only change is how existing out-of-fold rows are aggregated. Model, features, classifier states, probabilities, threshold, labels and folds are untouched. Before aggregating, the code asserts several things against the saved M2 artifacts. The row count is 27,558 and every path is unique. Folds, labels and Patient-IDs are identical to the saved assignment, and no Patient-ID spans two folds. Saved predictions equal $p \geq 0.5$. Pooled per-fold metrics recomputed with the M2 metric code match the saved ones to 1.1e-16 (tolerance 10^{-9}). Any failure raises an error.

For fold f with Patient-IDs P_f, let $P_f^+ \subseteq P_f$ hold those with at least one Parasitized cell. The pooled and macro sensitivities are

$$S_f^{\text{pool}} = \frac{\sum_{i \in P_f^+} TP_i}{\sum_{i \in P_f^+} (TP_i + FN_i)}, \tag{1}$$

$$S_f^{\text{macro}} = \frac{1}{|P_f^+|} \sum_{i \in P_f^+} \frac{TP_i}{TP_i + FN_i}, \tag{2}$$

and the primary quantity is $\Delta_f = 100\,(S_f^{\text{macro}} - S_f^{\text{pool}})$ pp. S_f^{pool} is the per-Patient-ID sensitivity weighted by each Patient-ID's Parasitized-cell count, so Δ_f isolates the change of weighting. Patient-IDs without Parasitized cells are counted per fold (10 / 10 / 10 / 9 / 11), not dropped. Specificity and accuracy are treated in the same way as secondary metrics. With $n = 5$ pairs, the smallest attainable two-sided exact Wilcoxon p is 0.0625. We therefore report effect sizes and intervals only, with no significance claims.

Ablations were planned in writing before they were computed and are exploratory; they vary only the aggregation. A1 checks the weighting identity. A2 requires at least $k \in \{1, 10, 25\}$ Parasitized cells per Patient-ID. A3 deletes one Patient-ID at a time. A4 is a cluster bootstrap over Patient-IDs within each fold (2,000 resamples, seed 42). A5 repeats the analysis for specificity and accuracy.

VII. RESULTS

Primary result. Table II and Fig. 2 give the five paired differences. 4 of 5 are negative; fold 2 is positive (+1.45 pp). The mean is −1.56 pp, inside the predicted −1 to −3 pp range, but 0.12 pp short of the −1.68 pp threshold. *Prediction vs. outcome:* the direction held in 4 folds and the magnitude fell inside the predicted range. The pre-registered threshold was not met, so the verdict is **INCONCLUSIVE**. Using the threshold's exact source value (1.679 pp) gives the same verdict. The 95% cluster-bootstrap CI for the mean is [−2.95, −0.04] pp. It excludes zero and includes −1.68 pp. The across-fold t-interval, [−4.18, 1.05] pp, is wider. Pooled M2 metrics are unchanged by construction.

Secondary metrics. Per-Patient-ID sensitivity varies more than per-Patient-ID accuracy in 4 of 5 folds: the SD across folds is 6.0 pp–13.4 pp for sensitivity vs. 3.6 pp–9.3 pp for accuracy (Fig. 1). Macro and pooled specificity almost coincide (mean +0.01 pp, CI [−0.03, 0.05] pp). The reason

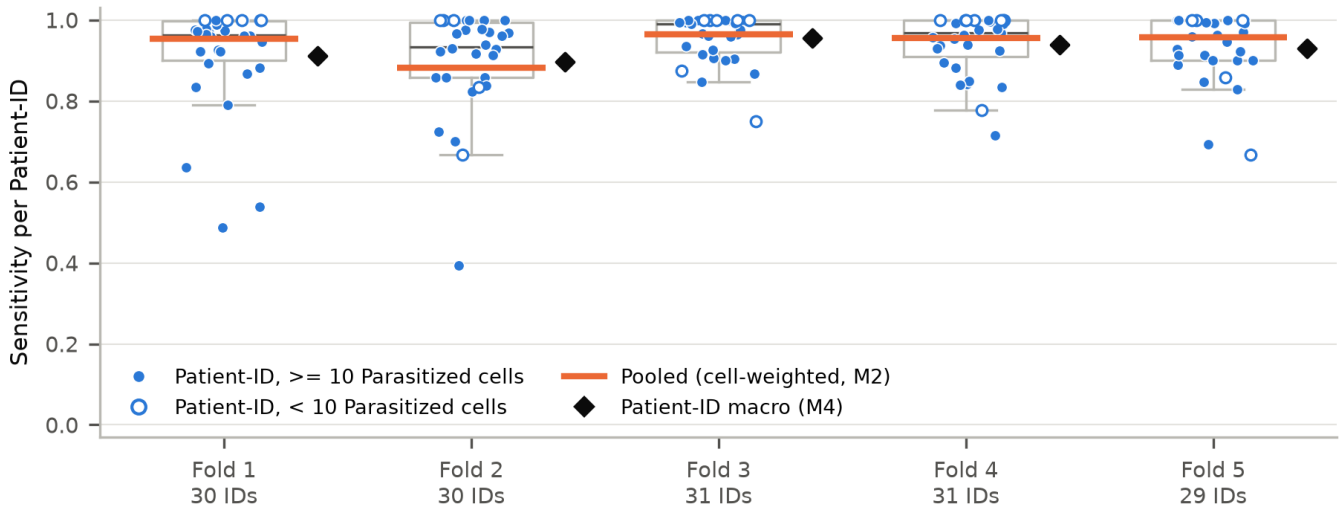


Fig. 1. Sensitivity of every Patient-ID with Parasitized cells, per test fold. Orange bar: pooled (cell-weighted, M2) sensitivity. Black diamond: Patient-ID-macro. Hollow markers: Patient-IDs with fewer than 10 Parasitized cells.

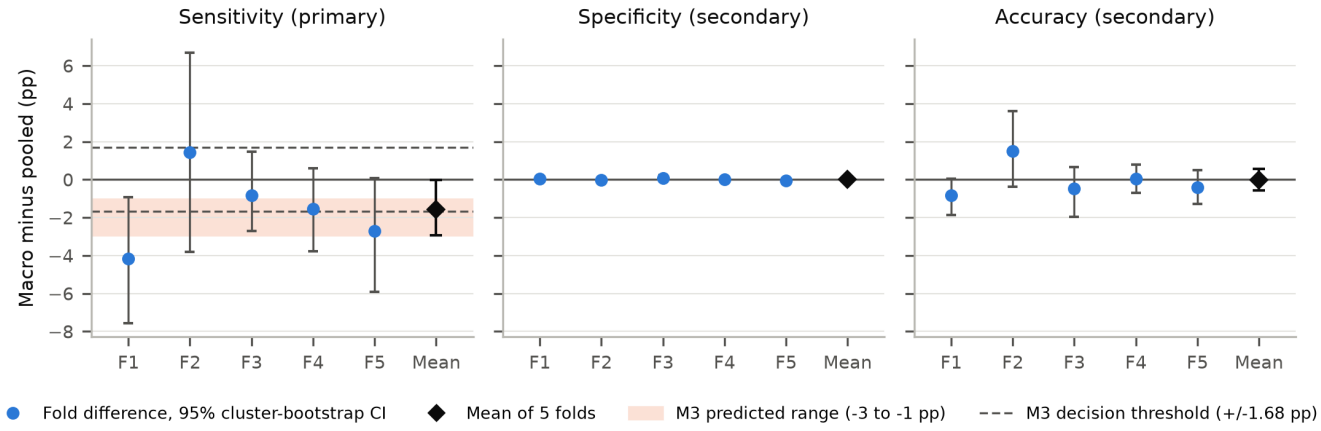


Fig. 2. Macro minus pooled per fold, with 95% cluster-bootstrap CIs (A4). The left panel is the pre-registered primary metric: the shaded band is the M3 prediction and the dashed lines are the ± 1.68 pp decision threshold. Specificity and accuracy are secondary.

TABLE II
PRIMARY RESULT: POOLED VS. PATIENT-ID-MACRO SENSITIVITY PER FOLD. VERDICT UNDER THE M3 RULE: INCONCLUSIVE.

Fold	IDs	Pooled (%)	Macro (%)	Δ (pp)	95% CI (pp)
1	30	95.31	91.15	−4.16	[−7.58, −0.93]
2	30	88.28	89.72	+1.45	[−3.81, 6.68]
3	31	96.38	95.55	−0.83	[−2.72, 1.48]
4	31	95.52	93.97	−1.55	[−3.79, 0.58]
5	29	95.71	92.99	−2.72	[−5.91, 0.08]
Mean	151	94.24	92.68	−1.56	[−2.95, −0.04]

is structural: every Patient-ID contributes 63–77 Uninfected cells, but between 1 and 633 Parasitized cells (median 33). Accuracy moves by −0.03 pp (CI [−0.57, 0.55] pp).

Ablations (exploratory). Table III summarises them. A1 confirms that cell-count weighting of per-Patient-ID sensitivity reproduces the pooled value exactly, in every fold. With A2's $k \geq 10$ eligibility, the mean becomes −2.00 pp, which would pass the threshold; with $k \geq 25$ it becomes −1.28 pp, which would not. Neither replaces the pre-registered $k \geq 1$ rule. Under A3, single-Patient-ID deletions keep the mean between −2.00 pp and −1.27 pp. 2 of them flip the verdict: C116P77ThinF (fold 2), C182P143NThinF (fold 5). The most influential is C116P77ThinF, with 414 Parasitized cells at sensitivity 72.5%. Removing it alone turns fold 2's difference to −0.73 pp.

VIII. FAILURE ANALYSIS

Where the errors are. The 795 false negatives occur in 106 of 151 eligible Patient-IDs. The top Patient-ID (C116P77ThinF) holds 14.3%, the top ten hold 51.4%, and 10 Patient-IDs account for half (Fig. 3). False positives are

TABLE III

EXPLORATORY ABLATIONS: AGGREGATION ONLY, MODEL UNTOUCHED. THEY DO NOT CHANGE THE PRIMARY VERDICT.

Ablation (exploratory)	Setting	IDs	Mean Δ (pp)	Folds < 0	Note
Primary (M3 rule)	macro, ≥ 1 Parasitized cell	151	−1.56	4/5	verdict INCONCLUSIVE
A1 weighting	per-ID, weighted by cells	151	–	-	identical to pooled in all folds (max abs. difference 0.0e+00)
A2 eligibility	≥ 10 Parasitized cells	121	−2.00	4/5	98.9% of positive cells kept
A2 eligibility	≥ 25 Parasitized cells	84	−1.28	4/5	94.5% of positive cells kept
A3 leave-one-ID-out	151 single-ID deletions	150 remain	–	-	mean range −2.00 to −1.27; 2 change the verdict
A4 cluster bootstrap	2,000 resamples, seed 42	151	−1.56	-	95% CI [−2.95, −0.04]
A5 specificity	macro over IDs with Uninfected cells	201	+0.01	3/5	95% CI [−0.03, 0.05]
A5 accuracy	macro over all IDs	201	−0.03	3/5	95% CI [−0.57, 0.55]

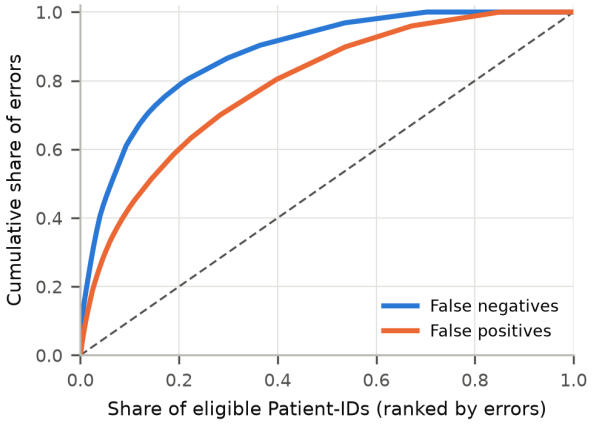


Fig. 3. Cumulative share of errors held by Patient-IDs ranked from most to fewest errors. Dashed: equal share.

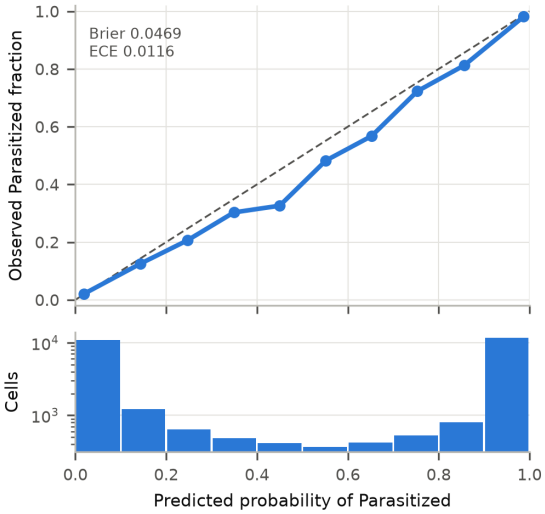


Fig. 4. Reliability diagram of the out-of-fold probabilities (10 equal-width bins) with bin counts (log scale).

more diffuse: the top ten hold 29.7%, and 28 Patient-IDs are needed for half. Patient-IDs with at least 100 Parasitized cells contribute 69.6% of false negatives. Fold 2 has the most false negatives (325) and contains 6 of the ten worst Patient-IDs. Its top Patient-ID alone holds 35.1% of the fold's false negatives. This explains the one positive fold. When large Patient-IDs are *hard*, cell weighting pulls pooled sensitivity below the macro, so the hypothesised masking runs in reverse. In the largest-SD fold (fold 1), no single Patient-ID drives the spread: removing the most influential one, C49P10thinF, cuts the SD by 18.6%.

Confidence and calibration. 228 false negatives have $p < 0.1$ (28.7% of all false negatives), and 217 false positives have $p > 0.9$ (24.5%). The Brier score is 0.0469 and the ECE 0.0116 (Fig. 4). Fold 2 is worst, with a Brier score of 0.0701. Mid-range bins over-estimate the Parasitized probability by up to 0.124.

Image review. One reviewer labelled 96 crops blind to class, probability and Patient-ID: 24 lowest-p false negatives, 24 highest-p false positives, and 24 random true positives and true negatives. The reviewer was the AI assistant used for the analysis, *not* a microscopist, and the categories were fixed before viewing (Table IV, Fig. 5). Among random correct cells, a stained inclusion was seen in 24/24 true positives and 0/24 true negatives. Among the most confident false negatives, 17/24 showed *no* visible inclusion; among the most confident false positives, 21/24 showed one. Grey/blue-tinged crops were more frequent among confident false negatives (20/24) than among random true positives (9/24). These are *observations, not causes*. They are consistent with label noise in the segmented crops [10], but equally with faint parasites, debris at the crop border, or reviewer error.

IX. DISCUSSION

The prediction was half right. Cell weighting does mask hard Patient-IDs in most folds, and the average effect is about the predicted size. Its direction, however, depends on whether a fold's large Patient-IDs happen to be easy or hard. With 30 / 30 / 31 / 31 / 29 eligible Patient-IDs in the five folds and Parasitized counts spanning two orders of magnitude, a few large Patient-IDs decide the sign. That is why one fold reverses and why single deletions flip the verdict. The practical lesson for reporting is that pooled and macro sensitivity answer different questions. A pooled value describes cells; a macro value describes the typical source group. Reporting both, with the per-group distribution and the count of groups with undefined sensitivity, shows variation that a single pooled number hides, such as a per-Patient-ID sensitivity of 39.4% (C50P11thinF, 33 Parasitized cells) in Fig. 1. The failure analysis suggests that some of the hardest "errors" may be label questions rather than model questions. Testing that would

TABLE IV

BLIND IMAGE REVIEW: COUNTS PER GROUP (ONE NON-EXPERT REVIEWER, CATEGORIES FIXED IN ADVANCE). "POSSIBLE LABEL NOISE" IS DERIVED BY CODE: THE VISIBLE CONTENT DISAGREES WITH THE DATASET LABEL.

Group (blind review)	Cells	IDs	Inclusion yes	No	Unclear	Grey/blue or atypical stain	Blur	Segmentation artefact	Possible label noise
Top FN (lowest p)	24	17	2	17	5	20	0	4	17
Top FP (highest p)	24	22	21	0	3	8	0	8	21
Random TP	24	19	24	0	0	9	1	8	0
Random TN	24	24	0	21	3	13	0	4	0

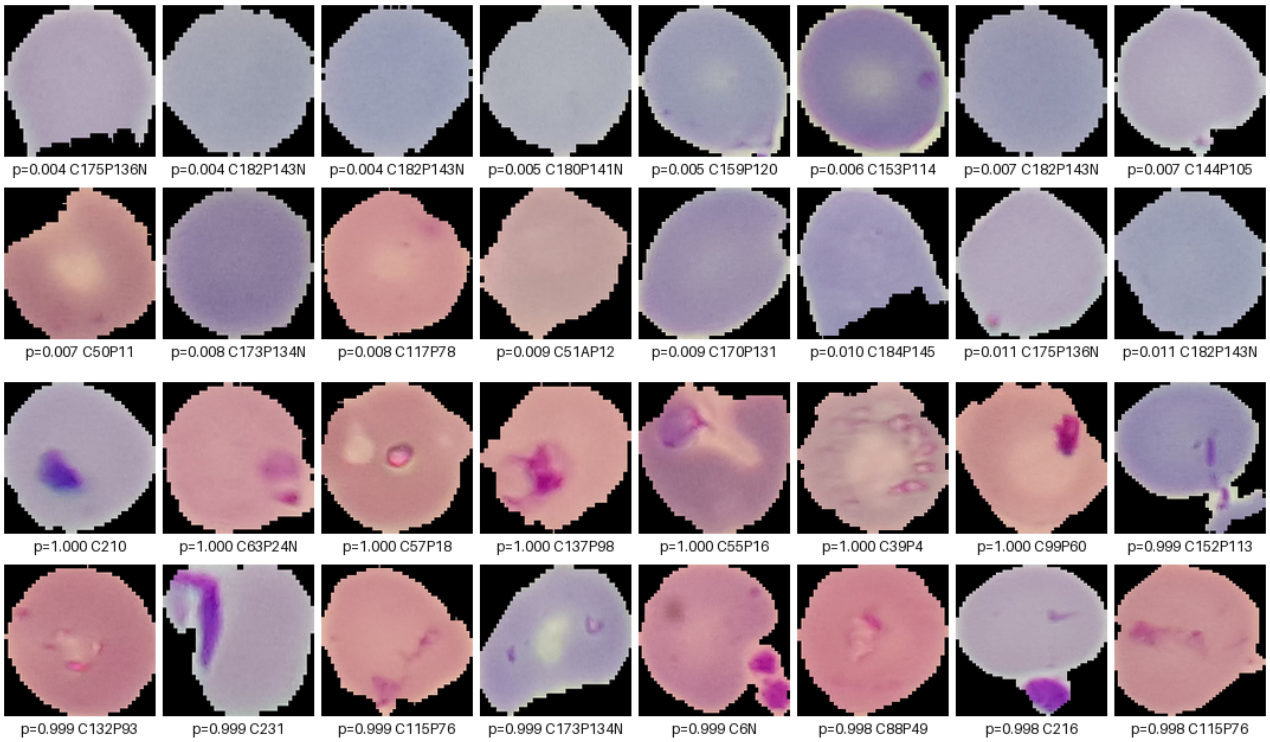


Fig. 5. Top: the 16 most confident false negatives (labelled Parasitized, lowest p). Bottom: the 16 most confident false positives (labelled Uninfected, highest p). Captions: p = P(Parasitized) and the Patient-ID prefix (display only). Crops are resized; colours are unaltered.

need expert re-annotation, which is outside this one-change study.

X. LIMITATIONS

- **Five folds, one fold assignment.** All paired statements rest on $n = 5$, so no significance can be claimed. Another grouping seed would regroup Patient-IDs and could change individual fold signs.
- **Fragile verdict.** The outcome is near the threshold: an eligibility change (A2) or a single deletion (A3) moves it. We report the pre-registered verdict and do not adopt the more favourable exploratory variants.
- **The Patient-ID is a mapping key, not a slide.** Uninfected crops include cells from infected slides, and one source patient has two microscope-specific keys. Results do not transfer to slide-level or patient-level diagnosis.
- **Approximate reproduction.** Folds, feature layer, normalisation and classifier differ from [6], and specificity is lower than published. The failure modes describe this model.
- **Image review.** One non-expert reviewer on 96 small crops, with no inter-rater agreement. The top-error groups were selected by model confidence.
- **Scope.** One public dataset of segmented *P. falciparum* thin-smear cells, with no external data. This is not a clinical or diagnostic claim.

XI. CONCLUSION

We reproduced a patient-level ResNet-50 malaria cell baseline approximately and pre-registered a single evaluation change. Patient-ID–macro sensitivity was lower than pooled sensitivity in 4 of 5 folds by -1.56 pp on average. That falls inside the predicted range but not past the pre-registered threshold, so the verdict is INCONCLUSIVE. The fold that reversed, and the fragility of the verdict, both follow from a few large Patient-IDs. Reporting pooled and per-group sensitivity side by side, with the group distribution, is a cheap way to make that structure visible. Next steps, each a separate intervention, are repeated grouping seeds, expert review of the confident errors, and stain-aware analysis.

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